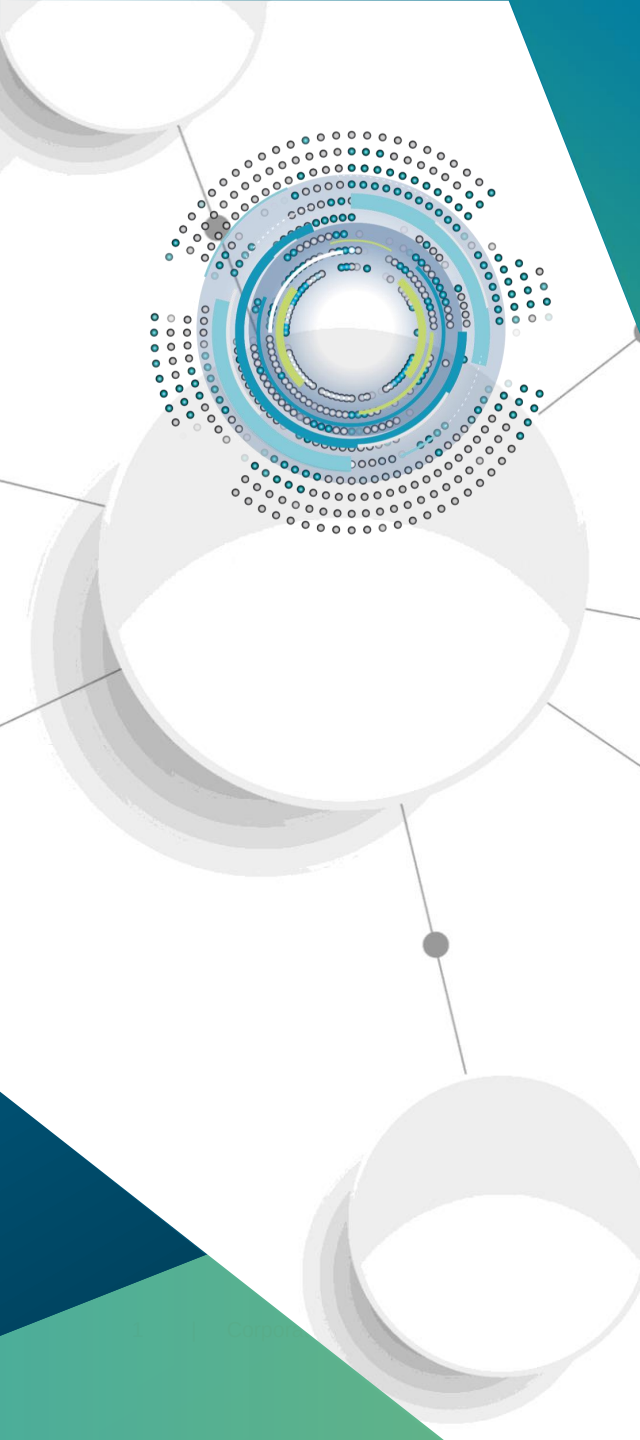




HEXAGON

empowering an autonomous future

MSCOne 2023

Your Key to a World of
Engineering Simulation

Unlock the World of Engineering Simulation

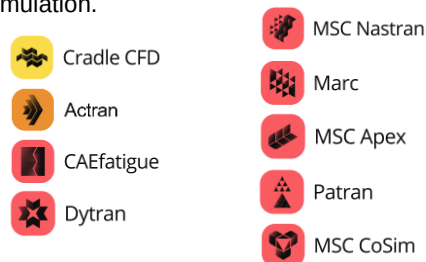
and more than meets the eye, with MSCOne



Multiphysics

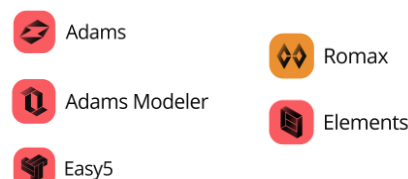
All physics simulations in one place

Create reliable and accurate digital twins with a complete suite of physics simulation software products including solutions for structural simulation and analysis, fatigue, computational fluid dynamics, acoustic and NVH simulation and multiphysics co-simulation.



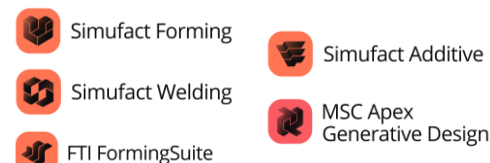
System Dynamics

Dynamic system simulation must consider complex multibody interactions. Hexagon's system dynamics solutions offer system-level insight into movement, enabling CAE-led development of complex products with reduced prototyping and enhanced collaboration through workflows with other Hexagon and third-party tools.



Virtual Manufacturing

Hexagon solutions simulate entire metal manufacturing process chains, including metal forming, welding and joining and additive manufacturing (AM). Our costing solutions establish target costs early in design to improve material utilization, reduce weight and minimize tooling development times.



Digital Materials

Empowered by synergistic integrated computational material engineering (ICME) and modelling, data management and artificial intelligence capabilities, Hexagon's materials autonomy solutions ensure the ability to meet the evolving needs of new markets and drive sustainable product design.



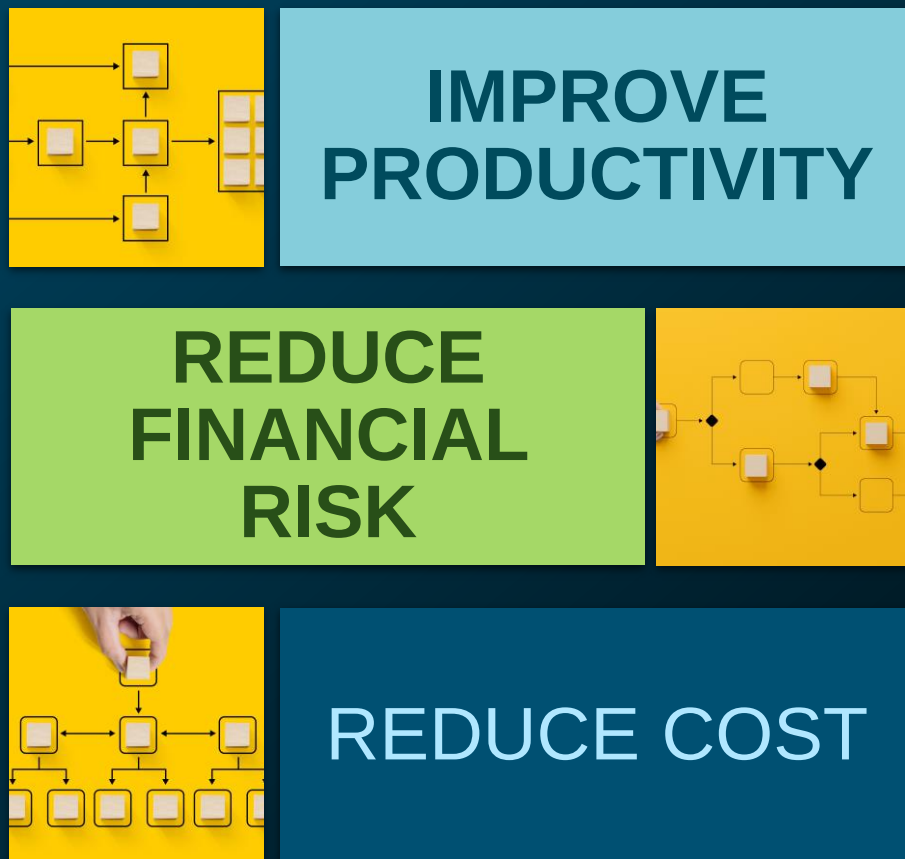
Simulation Governance



Artificial Intelligence & Machine Learning



Accelerate innovation with MSCOne



MSCOne

in Action

MSCOne Token Licensing

How to use tokens



MSCOne

Success Stories



**Increase agility
& collaboration
in engineering
development**

MSCOne

“MSC Nastran’s modules approach helps us save time and ensure our high-quality results... for NVH, Durability, and Nonlinear Analysis.”

- Dr.-Ing. Jürgen Bruns,
CAE-Methods, Volkswagen AG

Challenge

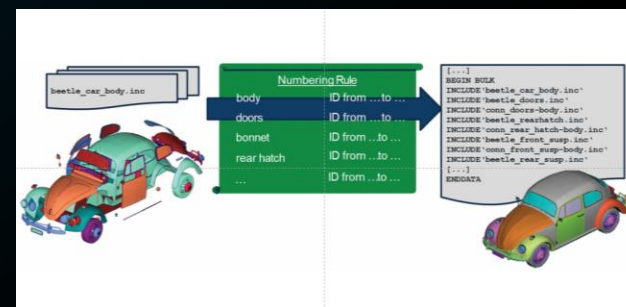
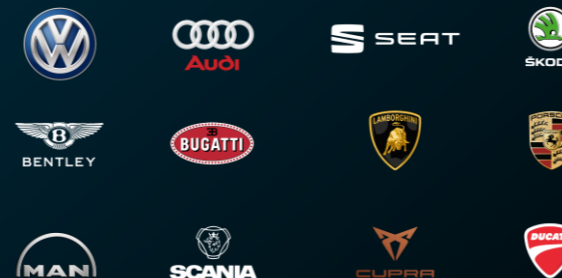
- VW announced a strategic alignment for eMobility
- Need to centralize software solutions, knowledge and experience while monitoring costs
- Expansion and optimization of their engineering/manufacturing workflows
- FEA model build-up contains multiple parts and substructures which need re-organization when merged for the complete vehicle

Solution

- Migrate existing Hexagon | MSC Software customer with Adams, MSC Nastran, Simufact, FTI, Actran, Digimat, VTD ... to MSCOne
- Centralize engineering software for efficiency, accuracy, performance, usage and costs
- MSC Nastran reduces reliance on pre-processing for creating copies of components and sub-systems when translating, rotating, or mirroring an existing MODULE – which reduces modeling time

Benefits

- Predictable costs for all engineering software
- Enable maximum flexibility when accessing physics-based solutions to explore variants in design
- Flexibility to enable expansion to new departments
- MSC Nastran Modules enable ease exchange of models with other analysis groups (such as statics, crash, NVH)





Reduce Need for Physical Prototyping with MSCOne

“The results show Adams-CFD co-simulation is predictive and that, as compared to the quasi-steady approach, has a superior correlation to the test data”

- Neil Lewington, Supervisor Aerodynamics
Ford Motor Company

Challenge

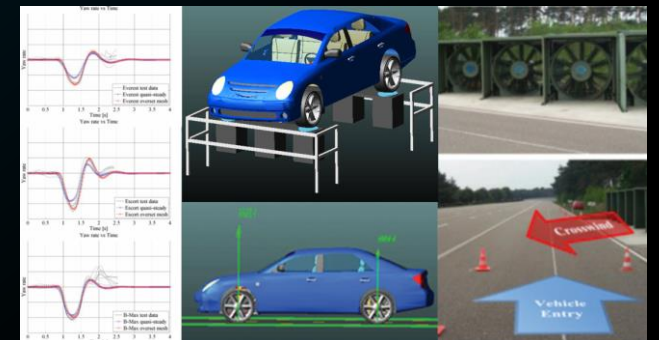
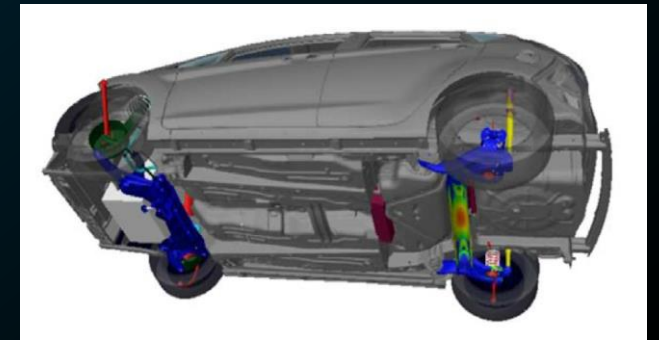
- Under constant pressure to address market demands in resource-constrained engineering environments
- Stiff development timelines are compounded by shrinking physical prototyping budgets
- Sufficient cross-wind stability essential criterion in a more controlled environment, than on-road testing to study off-normal events

Solution

- Fully coupled co-simulation between Adams and Ford’s CFD tool for high-fidelity modeling of severe events
- Full vehicle model in Adams Chassis and Overset mesh methodology used to drive the vehicles through the test setup
- Adams Chassis provides specialized environment to quickly build, test, refine, and optimize the performance a model of a full-vehicle

Benefits

- Early identification of aerodynamic issues that could adversely affect vehicle performance
- Elimination of high costs associated with design changes late in the design cycle
- Reduction in the level of physical prototyping required for vehicle sign-off





Accurately predict ride and handling performance with Machine Learning

MSCOne

“FDD modeling with ODYSSEE gave us the impetus to incorporate more complex damper characteristics and improve the fidelity of our simulation models”

- Lenka Visweswara Rao,
Mahindra & Mahindra

Challenge

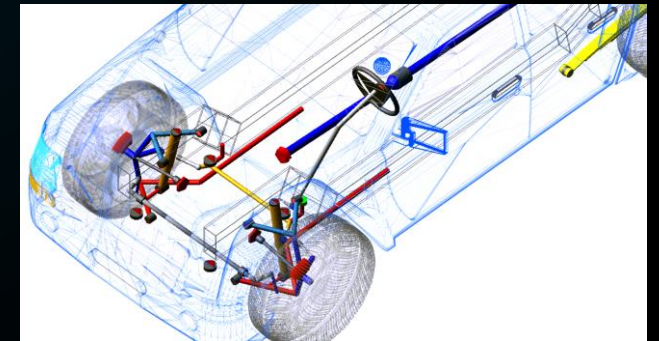
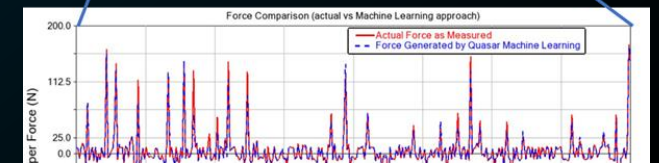
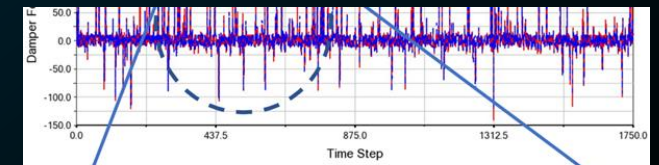
- Vehicle suspension systems require carefully design to balance between good ride comfort and good handling performance
- Frequency dependent damper (FDD) is used to get best-in-class ride and handling performance
- Pioneering effort to develop AI/ML models for modelling and predicting the performance of hydro-pneumatic struts in SUVs

Solution

- Adams Car used by engineering teams built and tested functional virtual prototypes of complete vehicles and vehicle subsystems
- Virtual vehicle designs variants were tested under various road conditions performing the same tests run in test labs or test tracks, but in a fraction of the time
- Incorporate suspension dampers in the Adams Car vehicle model and feed results to ODYSSEE for Machine Learning

Benefits

- Machine Learning model can replicate the real damper performance with high accuracy using Adams and ODYSSEE
- This task was nearly impossible without the ability to learn the test data via ODYSSEE CAE Quasar solver and its machine learning algorithms



AIRBUS

Simulation Helps Airbus Optimize Acoustic Liners & Reduce Aircraft Noise with MSCOne

"Actran is the only simulation tool able to accurately model the main physical phenomena for engine nacelle radiation"

- Jean-Yves Suratteau, Airbus

Challenge

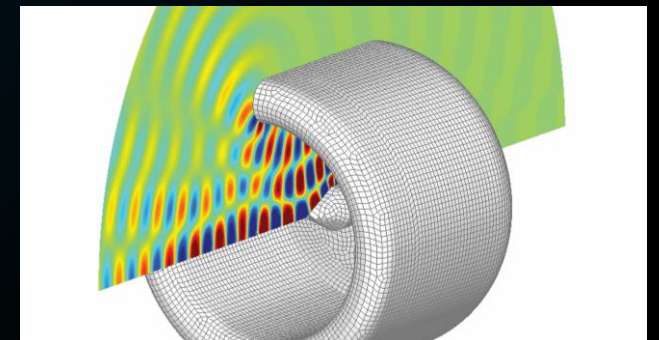
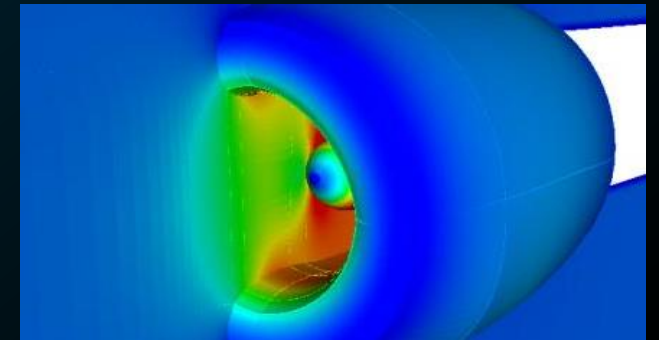
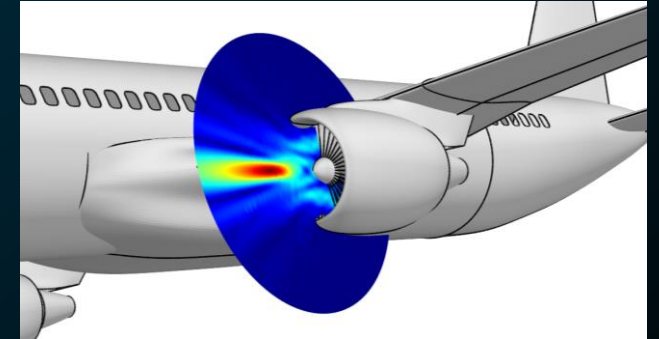
- Aviation noise is one of the most serious constraints on expansion and improvement of the air transport system throughout the world
- Turbofan engine fan noise is one of the largest contributors along with jet noise to noise radiated by an aircraft at take-off
- Airbus is working hard to reduce aircraft noise by improving the nacelle acoustic liners used to minimize the fan noise radiated from the engine

Solution

- Actran for acoustic design of nacelle liners in engine inlets and exhaust ducts, cockpit and cabin interior noise, air systems at ground, propellers and contrarotating propellers
- High-quality FE mesh of nacelle liner geometry with hexahedral and tetrahedral elements for mesh.

Benefits

- Dramatically reduced the time required to design and evaluate optimized acoustic liners by moving to a simulation-based process using Hexagon's| Actran acoustic simulation software





Virtual design and conditioning of progressive dies to shorten product development time

“With Simufact simulation we get robust progressive die process that accounts for variability during production. This allows us to get high quality part out the door faster”

- Dr. Soumya Subramonian,
Mfg & Process Development Engineer,
TE Connectivity

Challenge

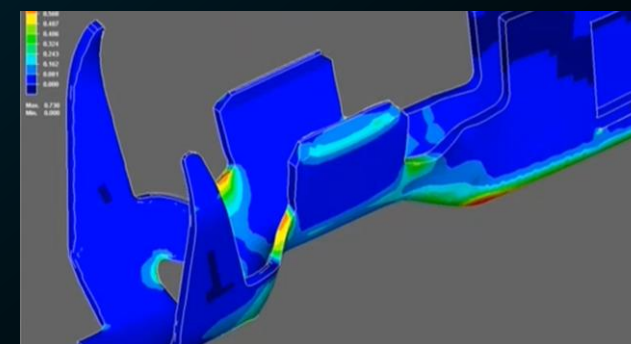
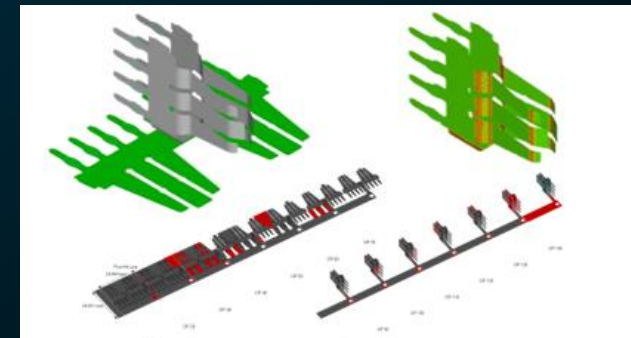
- Determine the material cost impact for new connector designs
- Reduce conditioning process to determine blank shape (2-4 weeks)
- Trial and error design process of electronics connector with a 30-stage forming process requires shorter development cycle
- Difficult or even impossible to control tolerances in the manufacturing
- Time-consuming and expensive development cycle

Solution

- FTI FormingSuite to develop accurate flat pattern and early feasibility studies based on part geometry identify potential formability concerns at design stage
- Simufact Forming simulates checks for material damage after forming and determines risk of cracking
- Compare simulated stamping deformations with CAD geometry at every step of the progressive die forming process
- Virtually design and optimise tooling geometry to identify & eliminate potential manufacturing issues (tool wear, tool stress)

Benefits

- Accelerated blank development process from two weeks to 2 minutes. Incredible productivity and accuracy gain (+/- 0.7%)
- Reduce tool wear and optimise tool design
- Enabled a global repeatable process where design engineers determine if a requested design is feasible early in the customer requirements process
- Save up to 50% material, time and money by replacing expensive and time-consuming physical tryouts with virtual tests.

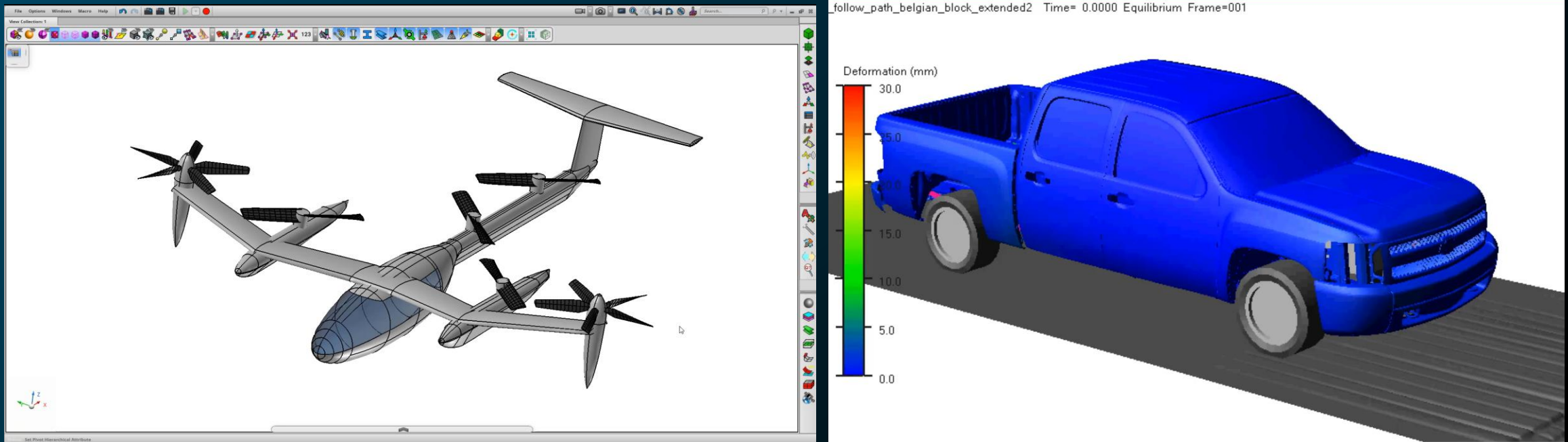


MSCOne

Showcases

Structural Simulation

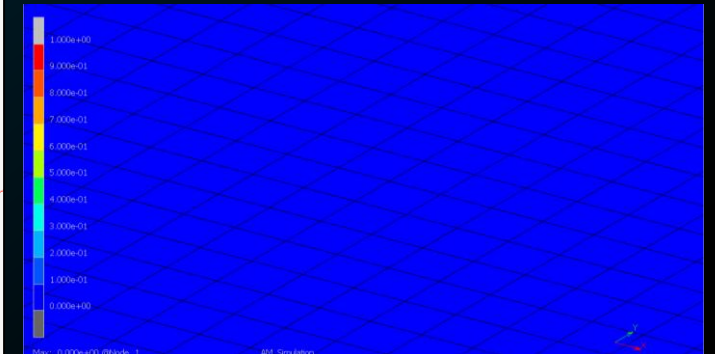
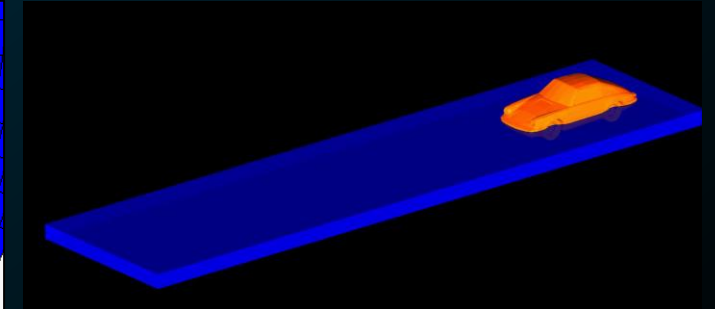
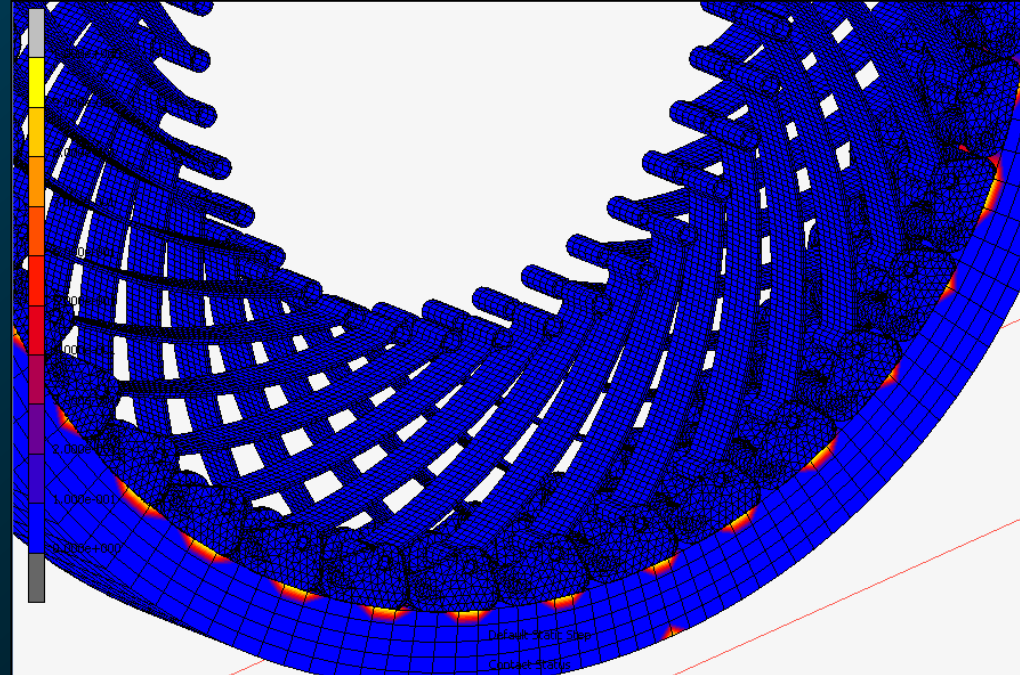
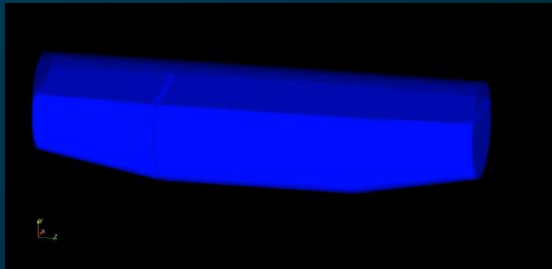
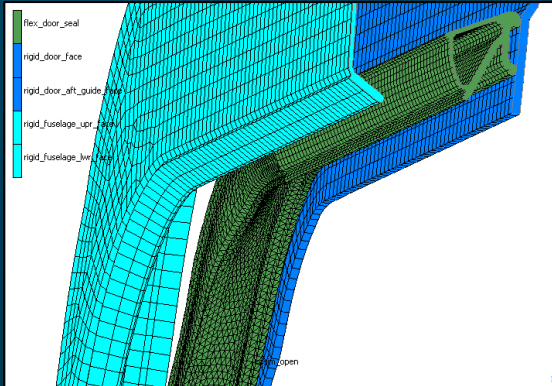
Delivering tried, trusted, industry standard, state-of-the-art structural analysis solutions



- MSC Nastran is the de-facto standard for aero-structure design, automotive NVH and the solver of choice for linear and mildly nonlinear problems
- CAEfatigue is state-of-the-art for durability analysis
- MSC Apex is a unified CAE environment for structural analysis, multi body dynamics and generative design
- We take structural simulation software to the next level in terms of accuracy, performance and user experience

Structural Simulation

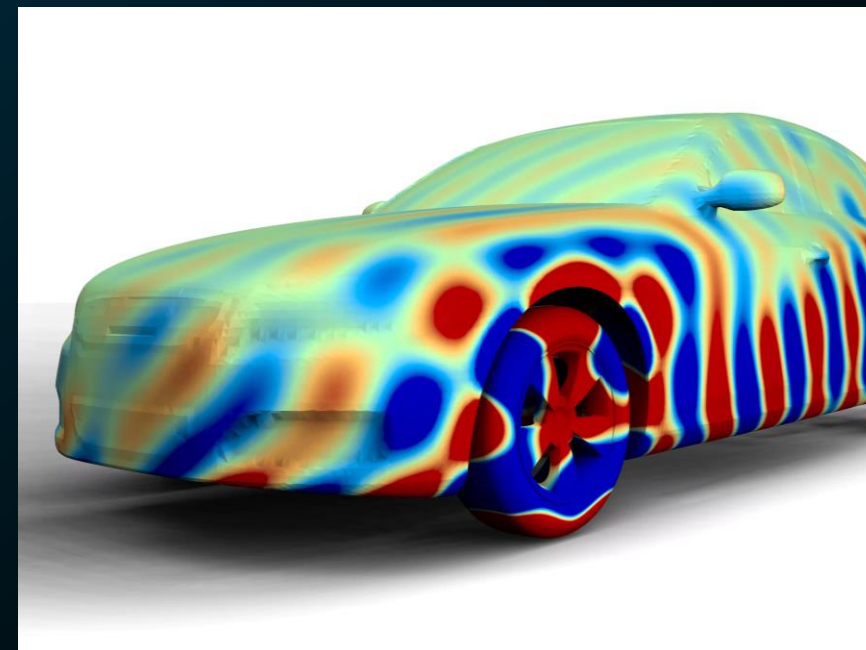
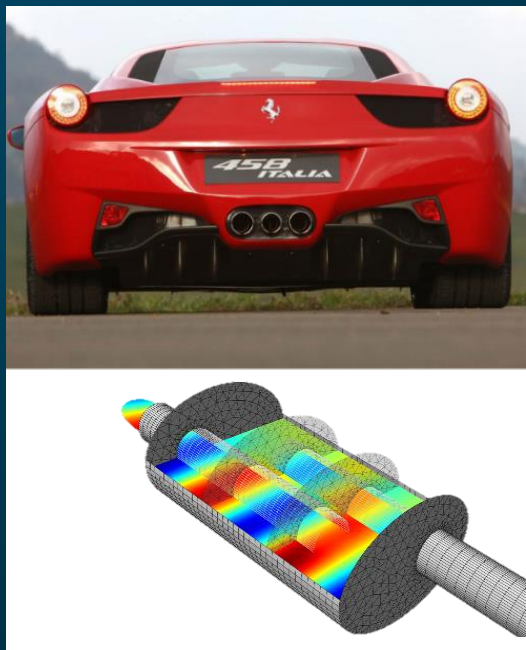
Delivering tried, trusted, industry standard, state-of-the-art structural analysis solutions



- Marc and Dytran are proven solvers for complex nonlinear problems
- We take structural simulation software to the next level in terms of accuracy, performance and user experience

Acoustics

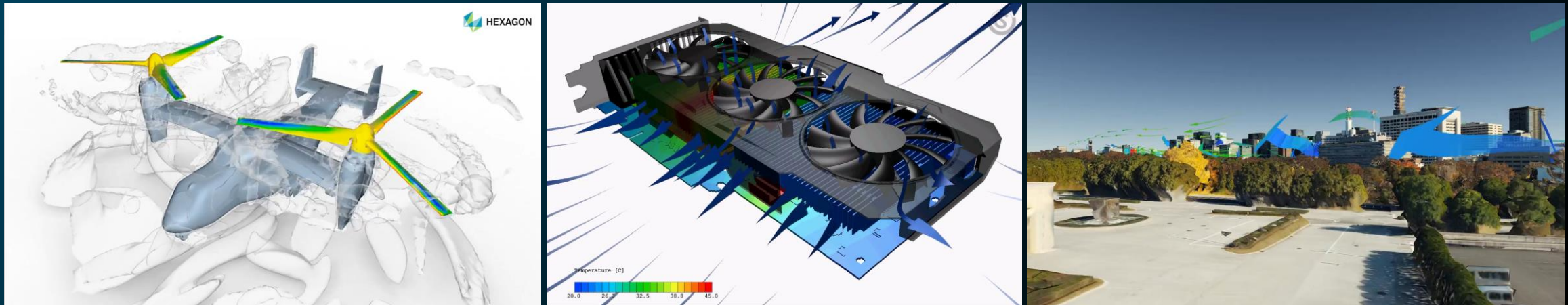
Solving the most challenging noise simulations in the world



- Actran is the premier solution for Vibro-Acoustics & Aero-Acoustics Simulation
- Actran helps automotive, aerospace and other customers across various industries comply with acoustics regulations, reduce noise pollution, improve comfort, and engineer sound for pleasure and brand recognition

CFD

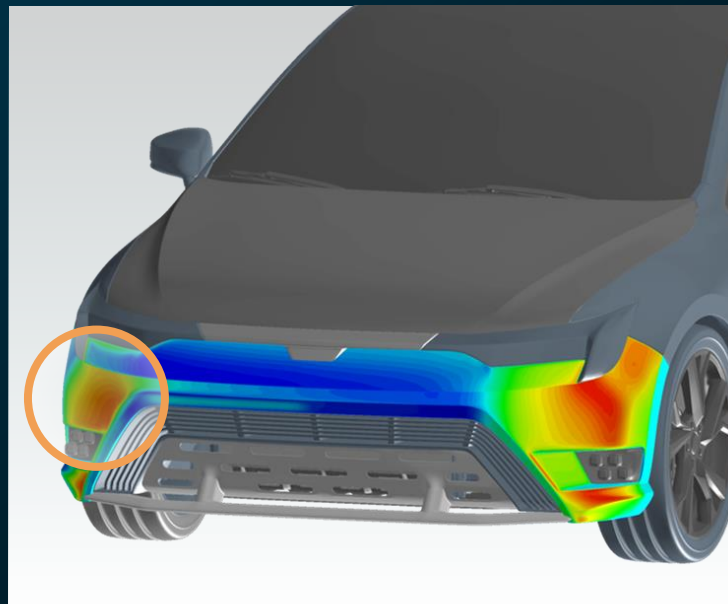
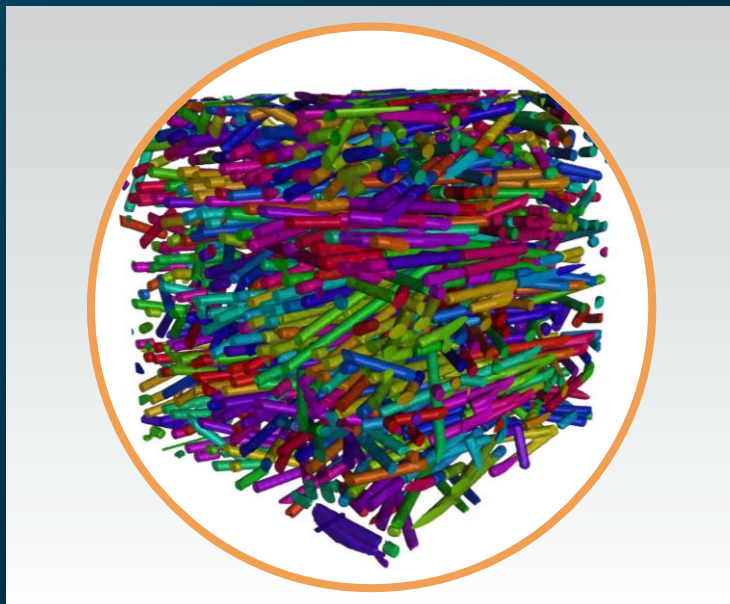
Delivering robust, fast, and accurate computational fluid dynamics solutions



- Robust, Fast & Accurate general purpose computational fluid dynamics software
- Unique solutions for Electronic cooling and the construction industry
- Strong capabilities for multi-physics co-simulation

Digital Materials

Delivering quality solutions for multi-scale material simulation & management

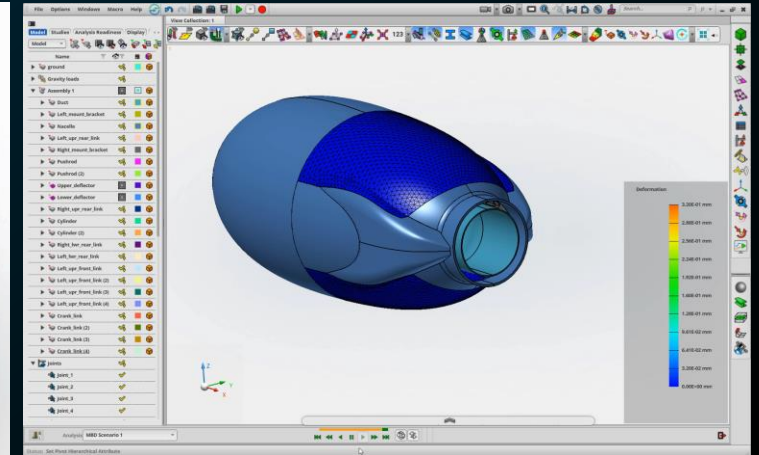
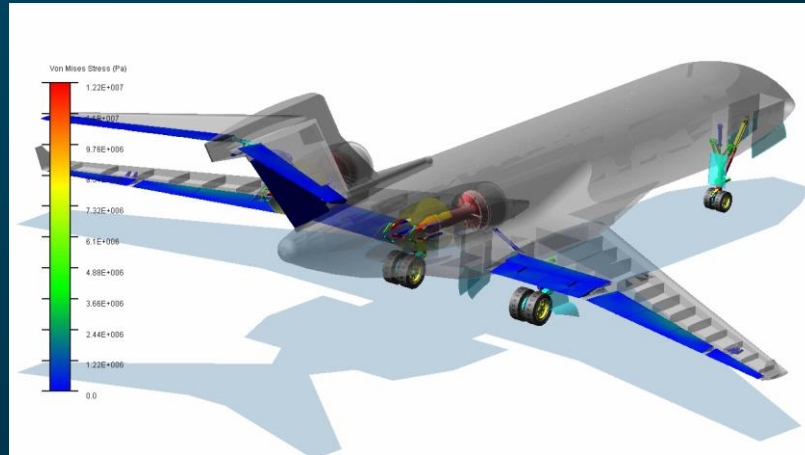


Release Level	Property Definition Label	Value	Category	Material Data
2-Production	Carbon Composition	[0.300 - 0.600] %	Composition	[Metals; Ferrous Metals; Ferrous Powder Metal Parts; Iron-NickelNickel St
2-Production	Copper Composition	[0 - 2.500] %	Composition	[Metals; Ferrous Metals; Ferrous Powder Metal Parts; Iron-NickelNickel St
2-Production	Iron Composition	[91.900 - 98.700] %	Composition	[Metals; Ferrous Metals; Ferrous Powder Metal Parts; Iron-NickelNickel St
2-Production	Tensile Elastic Modulus	1.5500E7 psi	Mechanical	[Metals; Ferrous Metals; Ferrous Powder Metal Parts; Iron-NickelNickel St
2-Production	Nickel Composition	[1.000 - 3.000] %	Composition	[Metals; Ferrous Metals; Ferrous Powder Metal Parts; Iron-NickelNickel St
2-Production	Yield Strength	25000 psi	Mechanical	[Metals; Ferrous Metals; Ferrous Powder Metal Parts; Iron-NickelNickel St
2-Production	Ultimate Strength	40000 psi	Mechanical	[Metals; Ferrous Metals; Ferrous Powder Metal Parts; Iron-NickelNickel St
2-Production	Poisson's Ratio	0.290	Mechanical	[Metals; Ferrous Metals; Ferrous Powder Metal Parts; Iron-NickelNickel St
2-Production	Strength	25000 psi	Mechanical	[Metals; Ferrous Metals; Ferrous Powder Metal Parts; Iron-NickelNickel St
2-Production	Density	6.1644E-4 slinches/in³	Physical	[Metals; Ferrous Metals; Ferrous Powder Metal Parts; Iron-NickelNickel St
2-Production	Tensile Ultimate Elongation	1.50E-6 in/in	Mechanical	[Metals; Ferrous Metals; Ferrous Powder Metal Parts; Iron-NickelNickel St

- Digimat is the state-of-the-art multiscale material modelling platform.
- Combined with material data management, manufacturing and structural simulation, we provide the most advanced ICME solution on the market.
- We empower our customers to leverage material data more effectively, increasing innovation and improving productivity.

System Dynamics

Solutions for Multibody dynamics, systems simulation & electro-mechanical drivelines



- Adams is the gold standard in multibody dynamics simulation
- Romax is the global leader in the design and analysis of gearboxes, drivetrains and bearings
- Elements expands the offering with multi-physics system integration, for rapid evaluation of performance and feasibility
- System dynamics is driving automotive, aerospace and other industries to develop next generation products with reduced physical prototyping and enhanced performance

Virtual Manufacturing

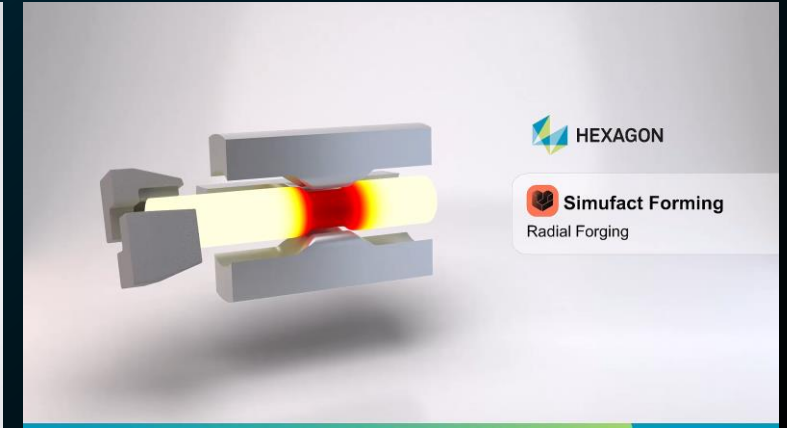
Quality Metal Forming, Virtual Assembly, Metal Additive Manufacturing and Costing Solutions



Design for Manufacturability



Design for Additive Manufacturing

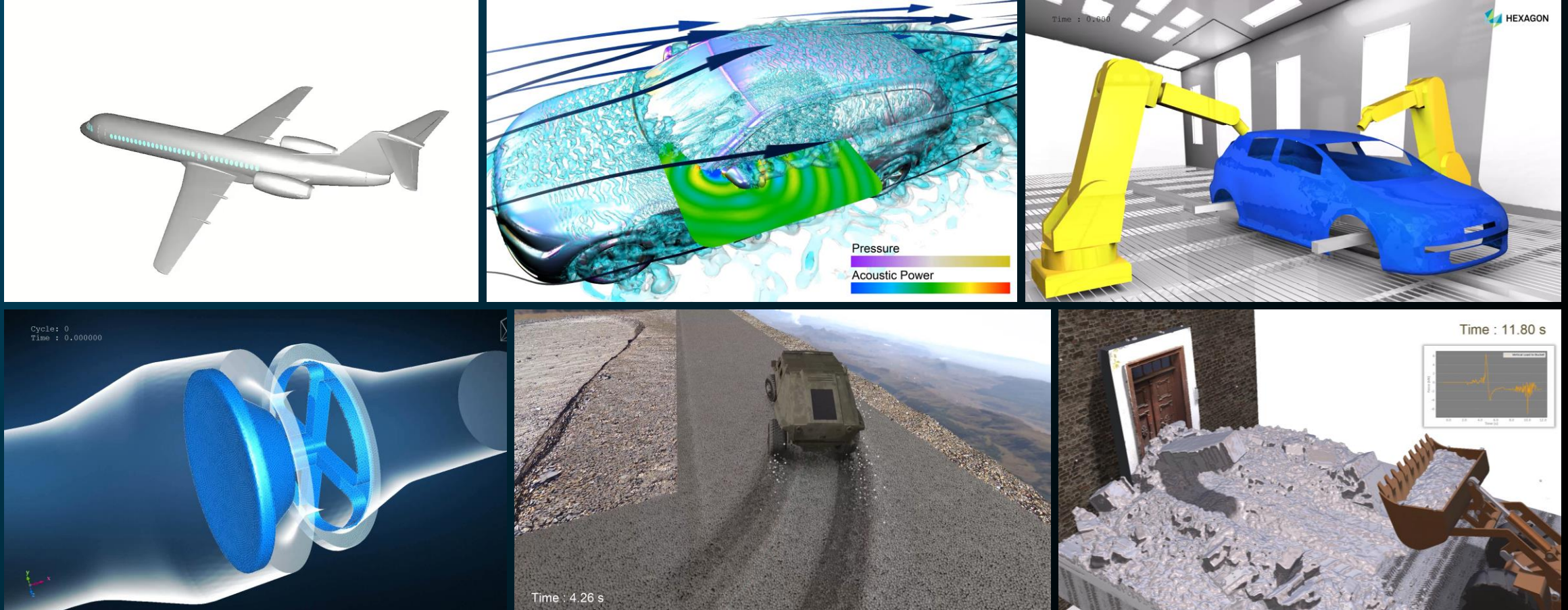


Bulk Forming

- Software for Virtual Manufacturing and Costing (VM&C) of metallic materials including Sheet Metal Forming, Bulk Forming, Virtual Assembly and the most advanced Additive Manufacturing Simulation on the market.
- Fast and highly automated Generative Design solution.
- VM&C software are built for accuracy, robustness and ease of use by shopfloor manufacturing engineers as well as CAE experts.

Co-simulation across the Software Portfolio

Fluids-structures-acoustics-motion all-in-one simulation for predictive accuracy



MSCOne

for You

MSCOne

Innovate, boost, adapt, optimize.



Enhance innovation by accessing new tools, boosting product development



Increase productivity and **adaptability** as project requirements evolve.



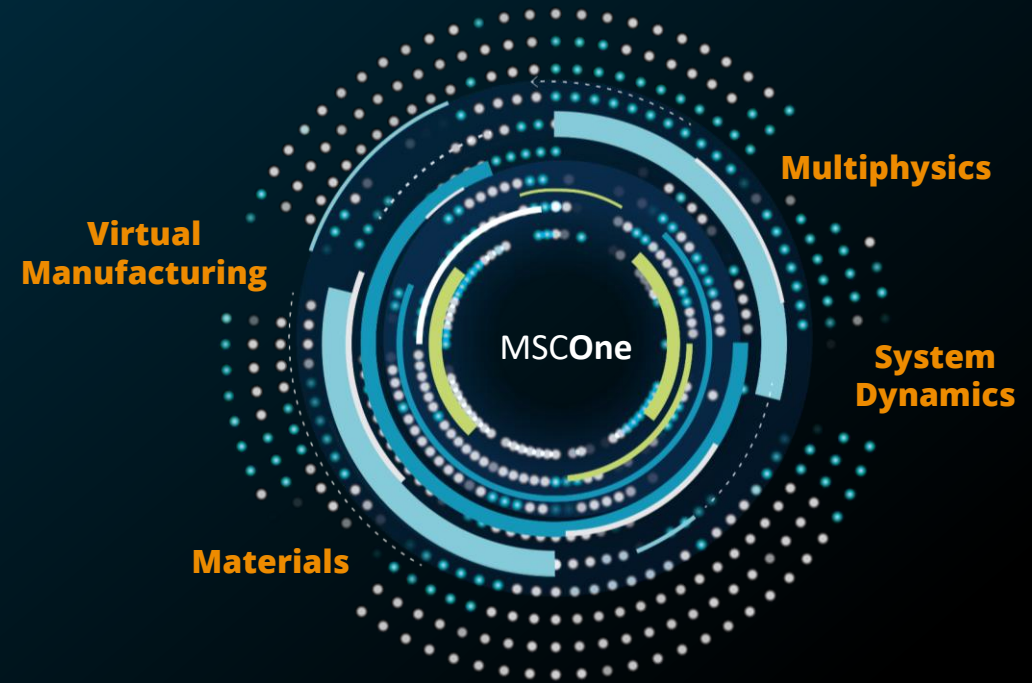
Mitigate financial risk by adjusting capacity to match engineering needs.



Save costs by utilizing specialist CAE applications efficiently, optimizing budget.



Accelerate knowledge cultivation with eLearning access.



MSCOne

Unleash the possibilities

